Week 2 Notes & Quiz

Notes for all six Week 2 lectures, made only from the 163 slides in *WEEK 2.pdf*. Each yellow tag such as S96 is the slide (PDF page) the point comes from. The 50-question practice questions and answer key are at the end.

STOP	LECTURE	SLIDES
L6	**E-Logistics**	S1–S28
L7	**International Logistics**	S29–S54
L8	**International Logistics: market penetration strategies**	S55–S86
L9	**Logistics Key Performance Indicators (KPIs)**	S87–S114
L10	**Tools for Improving Logistics Efficiency**	S115–S125
L11	**Tools for Improving Logistics Efficiency (contd.)**	S126–S163
+	**Cheat sheet · Slide quirks · Quiz (50 Qs)**	–

LECTURE 06 S1–S28

E-Logistics

Covers: introduction, the e-commerce scenario, a same-day delivery case, challenges, key elements S2

> E-LOGISTICS · EXACT WORDS
>
> *"E-logistics is defined as the management of all the physical flows of an organisation that sells goods on an online platform (website, marketplace, etc.)."* S3

- Also called **electronic logistics** or **digital logistics**: digital tools and systems used to manage and optimize logistics. S3
- It is set against traditional retailer logistics, but the two can be complementary. S3
- The withdrawal (return) period is much longer online, so e-merchants need a **returns management policy**. S4
- Fevad: 35% of customers with a bad delivery experience will not order from that site again. S4

The e-commerce scenario

CHART	WHAT IT SHOWS
Market concentration	**42%** of global e-commerce owned by 4 companies (2020). In India, 2 international companies control **63%** (2019): Flipkart 32%, Amazon 31%. S6
Penetration by category (China / US / India)	`Mobiles & electronics 70/60/30 · Fashion 70/40/10 · General merchandise 60/50/10 · Home/furniture 25/20/5 · Gr`
India segments 2021	Consumer electronics and apparel are the two biggest slices, about 40% each. S8
Festival season 2022	**₹40,000 crore** sold in the first week, 27% YoY growth. GMV share: Flipkart 62%, Amazon 26%. Order volume: Flipkart 49%, Meesho 21%. Sales vs usual: mobiles 7X, electronics 5 S9
Fashion e-commerce (Jan 2024)	Flipkart Fashion 27%, Myntra 22%, Meesho 15%, Amazon Fashion 15%, Ajio 4.5%, Nykaa Fashion 3%, others 13.5% S10

Solutions that matter to customers

- **Several delivery methods**: home delivery is most popular; pick-up points (Relay Point) are popular in France; express **D+1** delivery is almost unavoidable. Greener delivery: Ge French people interested, so bicycles and EVs for the last mile help stand out. S11
- **Free delivery**: PayPal/ComScore found **43%** of French online buyers abandoned a basket over high shipping costs. Alternatives: a minimum purchase amount, free delivery c delivery for a limited period. S12
- **Delivery tracking**: almost obligatory; reassures customers and saves customer-service time. S12

Key elements of e-logistics `10 items`

Supply chain visibility	Real-time tracking of goods and inventory; spot bottlenecks. S13
TMS	Transportation Management System: route optimization, carrier selection, load planning, shipment tracking. S13
WMS	Warehouse Management System: inventory, fulfilment, picking, packing, shipping. S13
Inventory management	Real-time tracking across locations; fewer stock-outs, better turnover. S14
EDI	Electronic Data Interchange: electronic exchange of purchase orders, invoices, shipping notices. S14

GTM	Global Trade Management: trade compliance, tariffs, customs documents, trade finance. S14
Last-mile delivery solutions	Route optimization, tracking apps, autonomous delivery vehicles. S15
Reverse logistics platforms	Returns, recalls, recycling. S15
Analytics & predictive insights	Data analytics and ML to anticipate demand and disruption. S16
Blockchain	Immutable, decentralized records for transparency and traceability. S16

Challenges: returns

- At least **30%** of online orders are returned vs **8.89%** in brick-and-mortar stores. 92% of consumers will buy again if returns are easy; 79% want free return shipping; 67% che 62% are more likely to shop online if they can return in-store; 49% of retailers offer free return shipping. S17
- Return reasons: wrong item 23%, product looks different 22%, damaged 20%, other 35%. S17
- Sample Amazon returns dashboard: ordered quantity 181,027, total returns 1,343, return rate **0.74%**. S18

Case: the same-day delivery race

Amazon's game plan	Warehouses in densely populated urban areas, inventory closer to customers. S19
Walmart's countermove	Pilot programs in select cities using its store network; partners with **UPS**; flat **$10** fee, no minimum order. S20
eBay's creative twist	Same-day delivery for items **not in its own inventory**, via local retailers and dedicated couriers. S21

- Asia Pacific same-day delivery market by type (B2B, B2C, C2C): 905.7 (2016) → 1,119.0 (2017) USD million, rising to 2027. S22
- Global same-day delivery: expected growth **18.8%** through 2027, market size **$15.8 bn** by 2027. S23
- Why same-day matters: 49% more likely to shop online if available; 51% of 18–34-year-olds expect it; 99% of retailers plan to offer it by 2024; it raises competitive edge by 85
- Quick commerce in India: **Blinkit 39%**, Swiggy Instamart 37%, Zepto 20%, others 4%; market about $700 million, expected $5.5 bn by 2025. S25
- What customers order (LocalCircles): last-minute essentials 37%, all groceries 29%, both 23%, indulgence items 11%. S26

LECTURE 07 S29–S54

International Logistics

Covers: international logistics, key components, cross-border transactions (types, steps, challenges), major stakeholders S30

> INTERNATIONAL LOGISTICS · EXACT WORDS
>
> *"...international logistics is the study, planning and implementation of how a business moves physical goods and materials from supplier to customer in a way that involves crossing at least one international border."* S31
>
> It also includes the international movement of **money and information**: business logistics spanning two or more countries. S31

Main goals 3 items

Cost efficiency	Usually the **first** goal: multiple tax regimes, tariffs, long-distance shipping. S32
Reliability	One misunderstood local rule can make a whole shipment unusable (confiscated, perishables turned away). S33
Relationships	Reliable, fair-priced delivery builds loyalty; can be the one competitive advantage for low-differentiation products. S33

Key components 9 items

Customs clearance	Commercial invoices, packing lists, certificates of origin, customs declarations; brokers or forwarders help. S35
Supply chain visibility	GPS, RFID, IoT sensors for real-time tracking. S35
Inventory management	Safety stock, demand forecasting, supply chain segmentation. S36
Risk management	Geopolitics, currency, trade barriers, disasters; diversify suppliers and routes, insurance. S36
Incoterms	**International Commercial Terms**: define delivery point, transport responsibility, transfer of risk between buyer and seller. S37
Cross-border regulations	Trade compliance, tariff classification, export controls, licensing. S37
Cultural & language considerations	Communication, cultural sensitivity, business etiquette. S38
Global supply chain network	Coordination among partners across countries. S38
Sustainability	Green transport, eco-friendly packaging, energy-efficient operations. S39

Cross-border transactions

- A financial transaction where the payer and recipient are in **separate countries**. Types: retail, wholesale, B2C, B2B. S40
- Value forecast to exceed $250 trillion by 2027, up over $100 trillion in a decade. S40
- Growth drivers: international trade and e-commerce; manufacturers' cross-border supply chains; global investment; remote workers paid across borders; an international workforce sending money home. S41

7 steps S42–S44

1. **Purchase or initiation** using local payment methods
2. **Routing and processing** via an encrypted gateway
3. **Payment verification** by the receiving bank (funds after conversion and fees)
4. **Transaction confirmation** (or a "declined" message)
5. **Transaction fulfilment**: merchant processes the order
6. **Settlement** through a global automated clearing house (**ACH**), typically **2–5 business days**
7. **Tracking**: each bank sends a reconciliation report

Cross-border payments can take weeks to reconcile and cost up to ten times a domestic transaction. S45 Challenges listed:

High funding costs	Capital set aside in advance in the right currencies. S45
Transaction fees	More entities, more bank fees. S46
Currency exchange rates	Rates move between initiation and settlement. S46
Long transaction chains	Each "link" adds time, fees, checks and data-corruption risk; multi-currency chains are longer. S46
Tax and legal implications	E.g. sales tax thresholds differ by country. S47
Limited operating hours	Settlement depends on banks' hours in both locations. S47
Fragmented, truncated data formats	Different standards; translation causes delays. S48
Legacy technology platforms	Batch processing, no real-time monitoring. S48
Complex compliance checks	Fraud, sanctions, financial-crime checks; false flags on similar names. S49

Major stakeholders in international consignments 10 items

Exporter	Seller or **shipper**; makes a sale to a foreign country. S51
Importer	The **buyer** purchasing from a foreign country. S51
Freight forwarder	Plans, coordinates and monitors all transport operations, including auxiliary services. S51
Customs agent	Authorized to do customs clearance formalities on others' behalf: documents, inspections, duties. S51
Shipping company	**Legal owner** of the vessel(s). S51
Shipping agent	Representative of the shipping company **at a given port**. S52

Consolidator	Sells cubic metres of container space; **groups goods from various exporters** to fill a container. Used by small-volume shippers. S52
Carrier	Moves the container from a location to the destination port. S52
Insurance	Covers incidents in transit for a premium. S52
Banks	Financial side, e.g. the **letter of credit**. S52

LECTURE 08 S55–S86

International Logistics: Market Penetration Strategies

Covers: exporting, licensing, joint ventures, ownership, importing, countertrade; large players; case study S56

The six strategies 6 items

STRATEGY	WHAT IT IS · TYPES	BENEFITS	EXAMPLES
Exporting	Sell goods made in one country to customers in another; **simplest and least risky**. Direct (no intermediaries) or indirect (trading companies, agents, distributors, export merchants). S58	Market diversification, competitive advantage, economies of scale, revenue growth S59	Tata Motors, Bajaj Auto (70+ countries), Infosys S60
Licensing	Licensor grants a licensee the right to use IP for **royalties**. Types: trademark, patent, copyright, **franchising**. S61–S62	Market expansion, risk mitigation, revenue, **low capital investment**, brand exposure S62	Amul (~20 countries), ITC (Aashirvaad, Sunfeast, Bingo), Fabindia S64
Joint venture	Strategic partnership, often across countries. **Equity JV**: new legal entity with shared ownership. **Contractual JV**: agreement, no separate entity. S65–S66	Risk sharing, access to resources, market entry, cost efficiency, risk diversification S66	Mahindra Susten, HUL–Unilever, Reliance energy JVs S68
Ownership	Wholly owned subsidiary (100%, **maximum control**, big investment), equity investments, franchising, acquisitions. S69–S70	Control and autonomy, market presence, knowledge transfer, risk management S71	Aditya Birla → Novelis, Tata Steel → Corus, Dr. Reddy's S73
Importing	Bring goods or services in from foreign sources for use, resale or distribution. S74	Product diversity, cost competitiveness, time-to-market, market expansion, risk mitigation S74	Walmart India (wholesale model), Amazon India, Apple India S76
Countertrade	A sale linked to a **reciprocal purchase**; beats forex shortages, currency restrictions, trade imbalances using non-monetary trade. S77	Market access, trade diversification, competitive advantage, risk mitigation S78	IOC (goods/services for crude from Saudi Arabia, Iraq), ONGC Videsh S80

Challenges by strategy

Exporting	Market entry barriers (tariffs), logistical, financial (exchange rates, defaults), legal & regulatory, cultural & language. S59
Licensing	Quality control (**no direct control**), IP protection, dependency on licensees, licensees becoming competitors, legal & contractual. S63
Joint venture	Cultural differences, strategic alignment, control & governance, IP protection, **exit strategies**. S66–S67
Ownership	**Capital requirements**, regulatory compliance, cultural differences, integration, political & economic risks. S71–S72
Importing	Trade barriers (tariffs, quotas), supply chain complexity, quality control, currency risk, competitive pressures. S74–S75
Countertrade	Complexity, negotiation, transaction costs, operational risks, financial risks. S78–S79

Large players

- Worldwide: XPO Logistics, UPS, NFI, DHL (Deutsche Post), Ryder, C.H. Robinson, DB Schenker, J.B. Hunt, Expeditors, FedEx. S81
- India: Lokesh Logistics, Blue Dart, Gati, FedEx, Mahindra Logistics, Safexpress, Gati-KWE, Allcargo, Delhivery, Ecom Express. S82

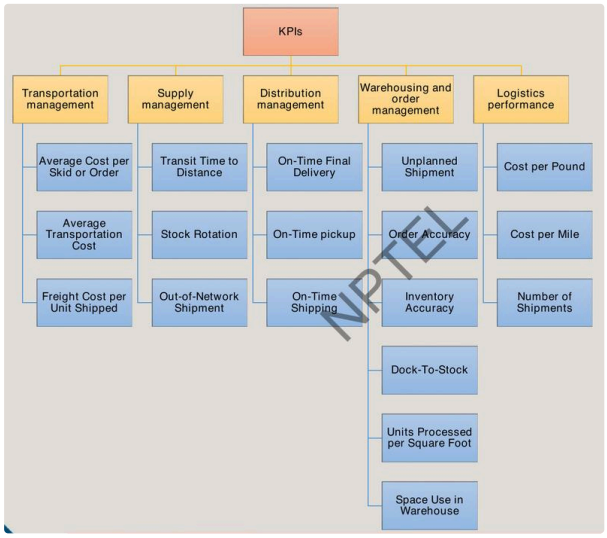
Case: syringes for Peru's COVID-19 vaccination drive

- **Challenge**: regular syringe shipments from a factory in China to Peru amid pandemic congestion and flight restrictions. S83
- **Strategy**: Noatum Logistics Peru found the only viable route, **48 hours** from the Shanghai factory. S83
- **Solution**: monthly **charter flights**, passenger area used for cargo; **8** air charters and part charters via Shanghai – Alaska – Los Angeles – Mexico – Lima. Peru's Foreign Ministry and Embassy in China helped get slots at Shanghai airport. S84
- **Result**: syringes arrived on time every month at reasonable cost. S84

LECTURE 09 S87–S114

Logistics Key Performance Indicators (KPIs)

- Measuring logistics performance means assessing efficiency, effectiveness and compliance with goals. KPIs include on-time delivery, order accuracy, inventory turnover, transport cost, warehouse utilization, customer satisfaction. S89
- Common KPIs: shipping time, order accuracy, picking accuracy, delivery time, pick & pack cycle time, order cycle time, equipment and trailer utilization, transportation costs, **dwell time** (time goods spend at a location), inventory carrying cost, warehousing cost, pick & pack cost, **operating ratio** (operating expenses vs revenue), packing material use, number of shipments, inventory accuracy, inventory turnover, inventory-to-sales ratio. S91–S93



Original slide S94. Five KPI groups: transportation, supply, distribution, warehousing & order management, logistics performance.

Formulas with the slide examples

KPI	FORMULA	SLIDE EXAMPLE → ANSWER
Transportation management		
Average cost per skid or order	Total costs ÷ number of orders	(500+1200+50+200+100) ÷ 300 = $6.83 S95
Average transportation cost	Avg monthly transport cost ÷ monthly gross income × 100	800 ÷ 10,000 × 100 = 8% S96
Freight cost per unit shipped	Freight cost ÷ units shipped	4,200 ÷ 6,000 = $0.70 S97
Supply management		
Transit time to distance	Pickup-to-destination time ÷ miles	5 days ÷ 2,000 = 0.0025 days/mile S98
Stock rotation (inventory turnover)	Sales ÷ average inventory	10,000 ÷ 5,000 → slide says 20 (the maths gives 2) S99
Out-of-network shipment	Out-of-network ÷ total shipments × 100	20 ÷ 5,450 × 100 = 0.4% S100
Distribution management		
On-time final delivery (OTD)	Delivered on time ÷ total × 100	15 ÷ 18 × 100 = 83% (goal 100%) S101
On-time pickup	Picked up on time ÷ total shipments × 100	43 ÷ 46 × 100 = 93.5% (>90% acceptable) S102
On-time shipping	Shipped on time and in full ÷ total shipped × 100	18 ÷ 23 × 100 = 78.3% S103
Warehousing and order management		
Unplanned shipment	Unplanned ÷ total shipments × 100	4 ÷ 4,000 × 100 = 0.1% S104
Order (picking) accuracy	Orders verified correct ÷ orders picked × 100	123 ÷ 128 × 100 = 96% S105
Inventory accuracy	Items counted ÷ items in system × 100	3,458 ÷ 3,506 × 100 = 98.6% S106
Dock-to-stock	Hours to put away ÷ shipments	12 ÷ 18 = 0.67 h/shipment S107
Units processed per sq ft	Units processed ÷ usable space	60,000 ÷ 16,400 = 3.7 S108
Space use in warehouse	Space with product ÷ total space × 100	3,300 ÷ 16,500 × 100 = 20% (average) S109
Logistics performance		
Cost per pound	Shipment cost ÷ weight (lb)	836.24 ÷ 2,200 = $0.38 (typical $0.22–$0.42) S110
Cost per mile (CPM)	(Fixed + variable costs) ÷ monthly miles	(2,540+2,312) ÷ 3,600 = $1.34 S111
Number of shipments	Monthly count, split by region or product	200 shipments: East 120, West 40, South 40 S112

WATCH THE DEFINITIONS

- OTD counts the **whole order**, not pieces S101; on-time shipping counts only **in-full** orders S103.
- On-time pickup: rely on documented transit time, not the carrier's data S102.
- Stock rotation is also called **inventory turnover** S99; order accuracy is also called **order picking accuracy** S105; space use is also called **space utilization** S109.

LECTURE 10 S115–S125

Tools for Improving Logistics Efficiency

- Quality in logistics = **conformity to requirements** and consistency in meeting standards. S117
- Quality assurance (QA) prevents mistakes and defects. Two principles: Fit for purpose and Right first time. S117
- Meaning of quality: the **producer's perspective** (production → quality of **conformance**: conformance to specifications, cost) and the **consumer's perspective** (marketing → quality of **design**: quality characteristics, price) both lead to **fitness for consumer use**. S118

DEMING'S COSTS OF QUALITY	EXAMPLES S119
Cost of conformance	Quality training, studies, surveys
Cost of non-conformance	Rework, scrap, inventory costs, warranty costs

Problem-solving techniques: **PDCA, 7 QC tools, quality circles, 5S, benchmarking**. S122

PDCA (plan-do-check-act)

- Also called the **continuous improvement spiral** or Deming wheel; drawn as a wheel because it is cyclical. S123
- Made popular by **William Edwards Deming**; its creation is attributed to his mentor **Walter Shewhart**. S123

Plan	Define the problem. Three sub-processes: identify the opportunity and analyse the start; set measurable targets; prepare an action plan. S123
Do	The **test phase**: perform the action, remove the causes. S124
Check	Evaluate results vs plan (often with data visualizations). Goal met → move to Act. Not met because the plan was poorly applied → go back and review deployment. S124
Act (adjust)	Document, standardize and monitor the change. S124

- In logistics, quality means meeting demand at the **lowest cost without compromising service level**. S125
- **Quality costs**: spent to obtain quality, e.g. a preventive maintenance plan. **Non-quality costs**: from lack of quality, e.g. downtime losses. S125

LECTURE 11 S126–S163

Tools for Improving Logistics Efficiency (contd.)

Covers: 7 QC tools, quality circles, 5S, benchmarking, conclusion S127

The seven QC tools

- Introduced by **Kaoru Ishikawa** in *Guide to Quality Control* (**1976**); these simple tools can solve 90% of quality problems. S128

TOOL	PURPOSE · SLIDE CASE
1. Flowchart	Shows steps and sequence; finds bottlenecks. Case: electronics assembly redesigned, training, quality checks, real-time monitoring. S129–S130
2. Check sheet	Easiest and fastest way to collect data as tally marks; mainly frequency data for histograms. Case: defects in electronic component assembly. S131–S134
3. Histogram	Shows variation (dispersion) and averages; classes on the horizontal axis, frequency on the vertical. Case: restaurant delivery times; the 25–30 min peak shrank, most orders moved to 15–20 min. S135–S137
4. Pareto diagram	**80-20 rule**: 80% of problems from 20% of causes; Ishikawa calls it the "first step to making improvements"; shows which problem to tackle first. Case: scratched screens (50) and battery issues (30) prioritized. S139–S141
5. Cause-and-effect	Also **fishbone** or **Ishikawa** diagram; finds root causes by category. Case: restaurant satisfaction; root causes understaffing and inefficient workflow. S142–S144
6. Scatter diagram	Relationship between two variables: positive, negative or no correlation. Case: more training programs → better job performance (positive). S145–S148
7. Control chart	Invented by **Walter A. Shewhart** in the 1920s (Shewhart chart); centre line = mean; upper and lower control limits at **±3 standard deviations**. Inside limits = common-cause variation (no action); outside = **special-cause** variation (investigate). S149–S152

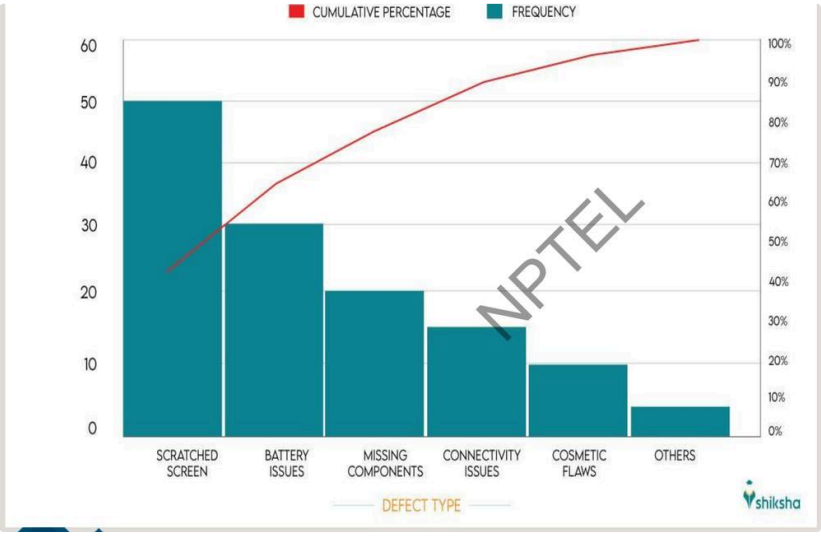
Name of the operator - **Date -**

Location - **Section -**

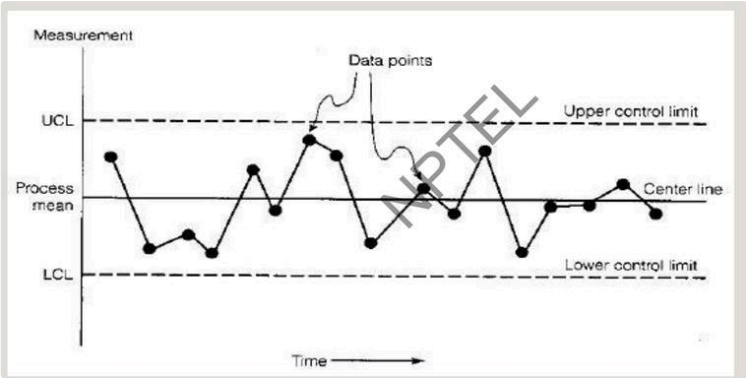
Defect Types	No of occurrences					Total
	Mon	Tue	Wed	Thu	Fri	
Bottles broken	II				III	5
Cap loose		II		I		3
Missing label	III		II		I	6
Dirt	I		II	II		5
Wrong order		III	I	II		6
Damage while packaging	II		II		II	6
Total	8	5	7	5	6	31

NPTEL

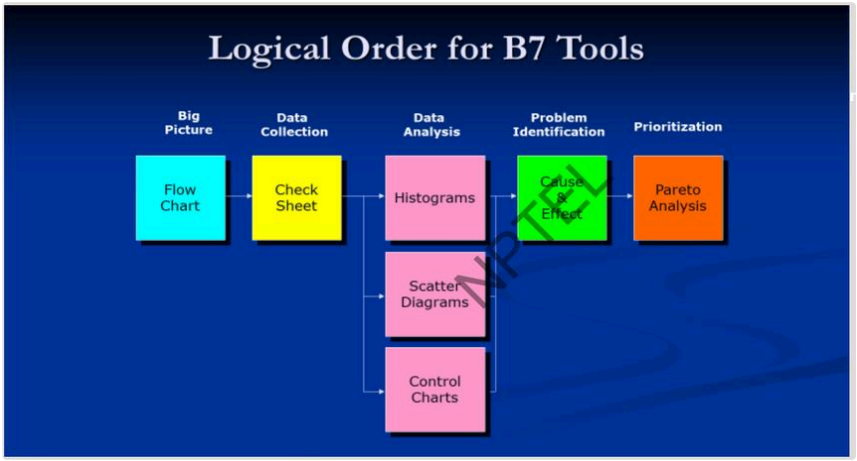
Original slide S132, check sheet example. Totals: missing label 6, wrong order 6, damage while packaging 6, bottles broken 5, dirt 5, cap loose 3; grand total **31**. Busiest day Monday (8).



Original slide S140, Pareto diagram: scratched screen 50, battery issues 30, missing components 20, connectivity 15, cosmetic flaws 10, others 5 (total 130). The top two make up 80 of 130, about 62%.



Original slide S150, control chart: measurement over time between the upper control limit (UCL) and lower control limit (LCL), around the process mean.



Original slide S153, logical order: big picture (flow chart) → data collection (check sheet) → data analysis (histograms, scatter diagrams, control charts) → problem identification (cause & effect) → prioritization (Pareto analysis).

Quality circles

- **Kaoru Ishikawa** is considered the father of quality circles. S154
- A **voluntary** group doing the same or similar work in the same area; meets **one hour a week** in normal working time; solves work problems at no or marginal extra cost. S154
- Size: ideally 8–10 members; never fewer than **5** or more than **15**. S154

Top management	Show support, give budget, set up awards, promote healthy competition. S155
Steering committee	Senior managers: framework, objectives, policy, **nominate coordinators and facilitators**, decide rewards. S155
Coordinator	Runs QC activity organization-wide; **registers all circles**; documentation; training material. S156
Facilitator	Senior officer of the department; arranges training and resources; **link between circles and management**. S156
Leader	Rotates; starts as a supervisor; convenes meetings, sets goals, delegates, builds consensus. S157
Members	From the work area; punctual, learn statistical tools, raise problems, contribute ideas. S157

5S philosophy

Five Japanese terms for waste reduction and an organized workplace. S158

Seiri · Sort	Keep only essential items.
Seiton · Set in order	Every item has a designated place.
Seiso · Shine	Keep areas clean and orderly.
Seiketsu · Standardize	Standards for organization and processes.
Shitsuke · Sustain	Keep practices going; audits for discipline.

Benchmarking

- A systematic process of measuring, comparing and improving performance against **industry leaders or best-in-class companies**. S159
- Six steps: define objectives → identify KPIs → gather data → select partners → analyse and compare data → identify opportunities. S160

CONCLUSION: SEVEN QUALITY CONCEPTS S161

Customer focus · continuous improvement (never-ending) · employee empowerment · use of quality tools · product design · process management (build quality into the process) · managing supplier quality.

BEFORE THE EXAM ALL LECTURES

Cheat sheet

NUMBER	MEANING	SLIDE
35%	Customers lost after one bad delivery (Fevad)	S4
43%	French buyers who abandoned baskets over shipping cost	S12
30% vs 8.89%	Online vs in-store return rates	S17
18.8% · $15.8 bn	Global same-day delivery growth rate · size by 2027	S23
39%	Blinkit's quick-commerce share	S25
$250 tn	Cross-border transactions by 2027	S40
2–5 days	ACH settlement time	S43
10×	Cross-border vs domestic transaction cost	S45
48 h · 8	Peru case: route time · charters	S83–S84
1976 · 90%	Ishikawa's QC tools book · problems they solve	S128
80-20	Pareto principle	S139
±3σ	Control limits	S151
8–10 (5–15)	Quality circle size	S154

3 goals	cost efficiency, reliability, relationships S32
6 entry strategies	exporting, licensing, joint ventures, ownership, importing, countertrade S57
7 payment steps	initiate → route → verify → confirm → fulfil → settle → track S42
5 KPI groups	transportation, supply, distribution, warehousing & order, logistics performance S94
PDCA	Plan, Do (test), Check, Act (adjust) S123
5S	Seiri, Seiton, Seiso, Seiketsu, Shitsuke S158

MEMORY HOOK (MY OWN, NOT FROM THE SLIDES)

QC tools in slide order: **F**low, **C**heck, **H**istogram, **P**areto, **C**ause-effect, **S**catter, **C**ontrol: "Fresh Coffee Helps People Concentrate, Study, Calmly".

READ ONCE SOURCE CHECK

Slide quirks to watch

- **Stock rotation maths:** $10,000 ÷ $5,000 = 2, but the slide prints "20". If the exam reuses these numbers, check which the options expect. S99
- **Cost per mile:** 4,852 ÷ 3,600 = 1.348, printed as $1.34 (truncated, not rounded). S111
- **Average transportation cost formula** is written as "Average transport cost = Average transport cost / (monthly gross income) × 100"; the first term should be average monthly transport cost. S96
- **Cross-border challenges** are numbered 1–6 then 8–10: there is no item 7. S45–S49
- **Franchising** is listed as a type of licensing (S62) and also as a type of ownership (S70). S62 S70
- Exporting benefit "Economies of Sale" means economies of scale; "Dr. Korau Ishikawa" on S154 is Kaoru Ishikawa. S59 S154
- Lecture 10's topic slide (S116) leaves out 5S, but S122 and S127 include it.
- Video or link-only slides: S5, S34, S50, S121. S90 and S120 are blank.

SELF-TEST 50 QUESTIONS

Week 2 practice questions

Answer these first, then check the key on the next pages. Option letters match the online quiz.

Q1 LECTURE 6

E-logistics is defined as the management of all the physical flows of an organisation that:

A owns a fleet of trucks
B runs bonded warehouses
C imports goods from abroad
D sells goods on an online platform

Q2 LECTURE 6

According to Fevad, what share of customers who had a negative delivery experience will no longer order on the same site?

A 43%
B 35%
C 25%
D 15%

Q3 LECTURE 6

Which key element of e-logistics allows electronic exchange of purchase orders, invoices and shipping notices between trading partners?

A EDI
B GTM
C WMS
D TMS

Q4 LECTURE 6

Global Trade Management (GTM) software mainly helps with:

A picking and packing
B route optimization
C trade compliance, tariffs, customs documentation and trade finance
D immutable record-keeping

Q5 LECTURE 6

Which e-logistics element offers 'immutable and decentralized record-keeping'?

A Reverse logistics platforms
B Analytics and predictive insights
C Supply chain visibility
D Blockchain technology

Q6 LECTURE 6

In Walmart's same-day delivery countermove, it partnered with ___ and charged ___.

A FedEx; free delivery above $35
B DHL; $5 per order
C UPS; a flat $10 fee with no minimum order
D USPS; a monthly subscription

Q7 LECTURE 6

What was distinctive about eBay's same-day delivery approach?

A Delivering items not stocked in its own inventory via local retailers and couriers
B Warehouses in dense urban areas
C Drone delivery only
D Free delivery for all orders

Q8 LECTURE 6

Which player leads India's quick-commerce market on the slide?

A BigBasket
B Swiggy Instamart
C Blinkit
D Zepto

Q9 LECTURE 6

Online orders are returned at a rate of at least 30%. What is the comparable brick-and-mortar figure on the slide?

A 8.89%
B 12%
C 27%
D 20%

Q10 LECTURE 7

International logistics involves moving goods from supplier to customer while crossing at least:

A one international border
B three countries
C two oceans
D one free-trade zone

Q11 LECTURE 7

Which of these is NOT one of the three main goals of international logistics?

A Cost efficiency
B Vertical integration
C Relationships
D Reliability

Q12 LECTURE 7

Incoterms are standardized terms that define:

A tax rates in each country
B shipping-line routes
C rights and responsibilities of buyers and sellers, such as delivery point and transfer of risk
D container sizes

Q13 LECTURE 7

The value of cross-border transactions is forecast to reach over ____ by 2027.

A $25 trillion
B $100 trillion
C $250 trillion
D $500 trillion

Q14 LECTURE 7

In cross-border transactions, settlement through a global automated clearing house (ACH) typically takes:

A the same day
B a few milliseconds
C two to five business days
D four to six weeks

Q15 LECTURE 7

Cross-border transactions can cost up to ____ more than a domestic transaction.

A twenty times
B ten times
C five times
D two times

Q16 LECTURE 7

Which stakeholder sells cubic metres of container space and groups goods from various exporters to fill a container?

A Shipping agent
B Customs agent
C Consolidator
D Carrier

Q17 LECTURE 7

The shipping agent is best described as:

A the bank issuing a letter of credit
B the legal owner of the vessel
C the representative of the shipping company at a given port
D the buyer of goods from abroad

Q18 LECTURE 8

Which market penetration strategy is described as one of the simplest and least risky?

A Joint venture
B Countertrade
C Ownership
D Exporting

Q19 LECTURE 8

Selling to foreign customers through trading companies, agents, distributors or export merchants is:

A licensing
B direct exporting
C indirect exporting
D countertrade

Q20 LECTURE 8

Which is a key challenge of licensing according to the slides?

A Very high capital requirement
B No direct control over quality
C Long settlement times
D Need to own vessels

Q21 LECTURE 8

In an equity joint venture, the partners:

A create a new legal entity in which each holds an ownership stake
B license trademarks for royalties
C exchange goods instead of money
D sign a contract without forming any new entity

Q22 LECTURE 8

Tata Steel's acquisition of Corus is the slide's example of which strategy?

A Countertrade
B Importing
C Licensing
D Ownership

Q23 LECTURE 8

Indian Oil Corporation trading goods or services for crude oil from Saudi Arabia and Iraq illustrates:

A Countertrade
B Franchising
C Indirect exporting
D Joint venture

Q24 LECTURE 8

Amul, ITC (Aashirvaad, Sunfeast, Bingo) and Fabindia are examples of:

A Ownership
B Countertrade
C Licensing
D Importing

Q25 LECTURE 8

In the Peru COVID-19 syringe case, how were the syringes moved?

A Monthly charter flights via Shanghai – Alaska – Los Angeles – Mexico – Lima
B By rail and road through Russia
C Through a 5PL aggregator's scheduled flights
D By sea via the Panama Canal

Q26 LECTURE 9

Bob's Hardware: marketing $500, COGS $1,200, packaging $50, shipping $200, storage $100, and 300 orders. Average cost per order is:

A $7.50
B $6.83
C $4.00
D $5.17

Q27 LECTURE 9

A company spends $800 a month on transport and has $10,000 monthly gross income. Average transportation cost is:

A 0.8%
B 12.5%
C 80%
D 8%

Q28 LECTURE 9

Freight cost of $4,200 for 6,000 units gives freight cost per unit shipped of:

A $1.43
B $0.70
C $0.07
D $7.00

Q29 LECTURE 9

A shipment takes 5 days to travel 2,000 miles. Transit time to distance is:

A 2.5 days/mile
B 0.025 days/mile
C 400 miles/day
D 0.0025 days/mile

Q30 LECTURE 9

Stock rotation is also known as:

A dwell time
B dock-to-stock
C space utilization
D inventory turnover

Q31 LECTURE 9

15 of 18 orders were delivered on time and in full. On-time final delivery (OTD) is about:

A 90%
B 83%
C 94%
D 78%

Q32 LECTURE 9

A carrier picked up 43 of 46 shipments on time. On-time pickup is ____; experts consider above ____ acceptable.

A 95%; 80%
B 93.5%; 90%
C 90%; 85%
D 93.5%; 99%

Q33 LECTURE 9

3,458 items are counted in stock while the system lists 3,506. Inventory accuracy is:

A 98.6%
B 101.4%
C 96.0%
D 99.2%

Q34 LECTURE 9

It took 12 hours to put away 18 shipments. Dock-to-stock time is:

A 0.15 hours per shipment
B 0.67 hours per shipment
C 6 hours per shipment
D 1.5 hours per shipment

Q35 LECTURE 9

Variable costs $2,540, fixed costs $2,312 and 3,600 miles a month give a cost per mile of about:

A $2.10
B $0.71
C $0.64
D $1.34

Q36 LECTURE 10

Quality assurance is based on which two principles?

A Fit for purpose, and Right first time
B Plan and Do
C Conformance and Design
D Sort and Sustain

Q37 LECTURE 10

In the 'Meaning of Quality' chart, quality of conformance belongs to the ____ perspective.

A consumer's
B producer's
C regulator's
D supplier's

Q38 LECTURE 10

Which is a cost of NON-conformance in Deming's costs of quality?

A Quality training
B Scrap
C Studies
D Surveys

Q39 LECTURE 10

PDCA was made popular by ____, though its creation is attributed to his mentor ____.

A W. Edwards Deming; Walter Shewhart
B Walter Shewhart; Joseph Juran
C W. Edwards Deming; Kaoru Ishikawa
D Kaoru Ishikawa; Walter Shewhart

Q40 LECTURE 10

Which PDCA phase is also known as the test phase?

A Act
B Plan
C Check
D Do

Q41 LECTURE 10

Losses due to machinery downtime are an example of:

A quality costs
B cost of conformance
C non-quality costs
D benchmarking costs

Q42 LECTURE 11

The 7 QC tools were introduced by Kaoru Ishikawa, who claimed they can solve about what share of a company's quality problems?

A 75%
B 50%
C 90%
D 80%

Q43 LECTURE 11

The Pareto diagram is based on which principle?

A ±3 sigma
B Right first time
C 80-20 rule
D Plan-do-check-act

Q44 LECTURE 11

Which QC tool is also called the fishbone or Ishikawa diagram?

A Scatter diagram
B Flowchart
C Control chart
D Cause-and-effect diagram

Q45 LECTURE 11

On a control chart, a point outside the control limits signals:

A a Pareto priority
B special-cause variation needing investigation
C common-cause variation needing no action
D a positive correlation

Q46 LECTURE 11

The control chart was invented in the 1920s by:

A W. Edwards Deming
B Kaoru Ishikawa
C Vilfredo Pareto
D Walter A. Shewhart

ANSWERS 50 QUESTIONS

Answer key

Q	ANS	WHY · SLIDE
Q1	D	S3: physical flows of an organisation that sells goods on an online platform (website, marketplace, etc.). S3
Q2	B	S4: 35%. (43% is the PayPal/ComScore basket-abandonment figure on S12.) S4
Q3	A	Electronic Data Interchange (EDI), S14. S14
Q4	C	S14: GTM handles international trade regulations, tariffs, customs documents and trade finance. S14
Q5	D	S16: blockchain reduces fraud risk through immutable, decentralized records. S16
Q6	C	S20: UPS partnership, flat $10 delivery fee, no minimum order. S20
Q7	A	S21: eBay offered same-day delivery for items not in its inventory through local retailers and dedicated couriers. S21
Q8	C	Blinkit 39%, Instamart 37%, Zepto 20%, others 4%. S25
Q9	A	S17: at least 30% online vs 8.89% in brick-and-mortar stores. S17
Q10	A	S31 definition; it also includes international movement of money and information. S31
Q11	B	The goals are reliability, relationships and cost efficiency (usually the first). S32
Q12	C	S37: International Commercial Terms define delivery point, transport responsibility and transfer of risk. S37
Q13	C	S40: over $250 trillion by 2027, up more than $100 trillion in a decade. S40
Q14	C	S43: settlement typically takes two to five business days. S43
Q15	B	S45: up to ten times more, and can take weeks to reconcile. S45
Q16	C	S52: the consolidator. Small-volume shippers usually hire one. S52
Q17	C	S52. The shipping company (S51) is the legal owner of the vessel. S52
Q18	D	S58: exporting uses existing products and capabilities to reach new customers abroad. S58
Q19	C	S58: indirect exporting uses intermediaries; direct exporting does not. S58
Q20	B	S63: Quality control, no direct control. Licensing itself needs low capital investment (S62). S63
Q21	A	S65: equity JV creates a new legal entity; a contractual JV (S66) does not. S65
Q22	D	S73: ownership examples are Aditya Birla–Novelis, Tata Steel–Corus and Dr. Reddy's. S73
Q23	A	S80: countertrade links a sale to a reciprocal purchase. S80
Q24	C	S64: licensing examples. S64
Q25	A	S84: 8 air charters and part charters along that route, using passenger space for cargo. S84
Q26	B	(500+1200+50+200+100)/300 = 2,050/300 = $6.83. S95
Q27	D	800 / 10,000 × 100 = 8%. S96
Q28	B	4,200 / 6,000 = $0.70 per unit. S97
Q29	D	5 / 2,000 = 0.0025 days per mile. S98
Q30	D	S99: stock rotation = inventory turnover = sales / average inventory. S99
Q31	B	15 / 18 × 100 = 83%. The goal is always 100%. S101
Q32	B	43 / 46 × 100 = 93.5%; anything above 90% is acceptable. S102
Q33	A	3,458 / 3,506 × 100 = 98.6%. S106
Q34	B	12 / 18 = 0.67 hours per shipment. S107
Q35	D	(2,540 + 2,312) / 3,600 = $1.34 per mile as printed (1.348). S111
Q36	A	S117: fit for purpose; right first time (eliminate mistakes before they happen). S117
Q37	B	Producer's perspective → production → quality of conformance (conformance to specs, cost). Consumer's perspective → quality of design. S118
Q38	B	Non-conformance: rework, scrap, inventory costs, warranty costs. Conformance: training, studies, surveys. S119
Q39	A	S123: popularized by Deming, created by his mentor Shewhart. S123
Q40	D	S124: Do is the test phase. S124
Q41	C	S125: non-quality costs come from a lack of quality. Preventive maintenance is a quality cost. S125
Q42	C	S128: 90%, from Guide to Quality Control (1976). S128
Q43	C	80% of problems come from 20% of causes. S139
Q44	D	S142: the cause-and-effect diagram. S142
Q45	B	S151: limits are set at ±3 standard deviations; outside points indicate special-cause variation. S151
Q46	D	S149: it is also known as the Shewhart chart. S149
Q47	D	Big picture: flow chart → data collection: check sheet → data analysis → problem identification: cause & effect → prioritization: Pareto. S153
Q48	B	S154: ideally 8–10; never fewer than 5 or more than 15. S154
Q49	B	S156: the coordinator registers circles, organizes documentation and training material. S156

Q	ANS	WHY · SLIDE
Q50	**B**	Seiri = sort, Seiton = set in order, Seiso = shine, Seiketsu = standardize, Shitsuke = sustain. S158

Made from *WEEK 2.pdf* (163 slides, NPTEL Logistics & Supply Chain Management, Prof. Vikas Thakur, IIT Kharagpur). Notes use only the slides, except boxes marked "my own".